THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS IX — SCIENCE (086) · MID-TERM / HALF-YEARLY PRACTICE PAPER

Session 2026-27 · School Examination Pattern

Class: 9	Subject: Science	Set 04	Time: 3 Hours	Max. Marks: 80

General Instructions:

1. This question paper consists of 39 questions in 3 sections. Section A is Biology (Q.1–16), Section B is Chemistry (Q.17–27) and Section C is Physics (Q.28–39).
2. All sections are compulsory. However, internal choice is provided in some questions. A student is expected to attempt only one of the options in such questions.
3. Section A carries 32 marks, Section B 25 marks and Section C 23 marks. Marks are printed against each question.
4. Draw neat, labelled diagrams wherever required. Use g = 10 m s^{-2} wherever needed.

SECTION A — BIOLOGY — 16 questions · 32 marks

Q.1–7 MCQs (1 mark each); Q.8–9 Assertion–Reason. For Assertion–Reason: (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

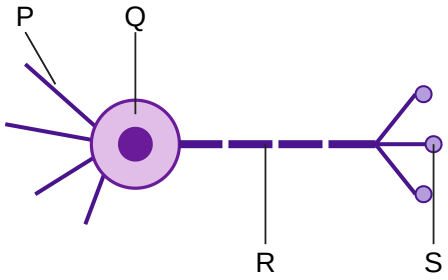
1. Which cell organelle contains its own DNA and ribosomes and can therefore make some of its own proteins? **1**

(a) Golgi apparatus (b) Mitochondria
(c) Lysosome (d) Endoplasmic reticulum

2. A cube of potato kept in a strong sugar solution for an hour loses mass. This is because: **1**

(a) the sugar solution is hypotonic to the potato cells
(b) the potato cells are hypertonic to the solution
(c) water moves out of the potato cells by osmosis since the solution is hypertonic
(d) sugar enters the potato cells by osmosis

3. In the neuron shown, the part that carries the nerve impulse away from the cell body is: **1**



(a) P (b) Q (c) R (d) S

4. A muscle has spindle-shaped cells, each with a single nucleus, no striations and works involuntarily. It is: **1**

(a) skeletal muscle (b) cardiac muscle
(c) smooth muscle (d) striated muscle

5. Which of the following would NOT help in reducing the enhanced greenhouse effect? **1**

(a) planting more trees (b) using solar energy
(c) burning more coal (d) using public transport

6. Which of the following is a non-flowering seed plant (gymnosperm)? **1**

(a) Fern (b) Pine (c) Moss (d) Mango

7. Which of the following is NOT a method of vegetative propagation? **1**

(a) Grafting (b) Layering (c) Pollination (d) Cutting

8. **Assertion (A):** Xylem transports water and minerals from the roots to the leaves. **1**
Reason (R): Xylem vessels and tracheids are dead cells with lignified walls that form continuous tubes.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** Insolation at the Earth's surface is always equal to the solar constant. **1**
Reason (R): Part of the solar radiation is absorbed and reflected by the atmosphere.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. Why is the cell membrane called selectively permeable? **2**

OR

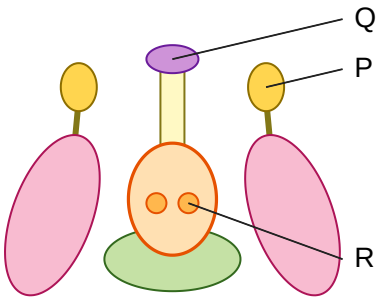
What are plastids? Name two types of plastids.

11. Name the tissue that (i) stores fat in the body, (ii) forms the outer covering of the skin and (iii) carries oxygen in the blood. **3**

OR

What is cartilage? Name two places where it is found in the body and state one function.

12. The figure shows a flower. Name the parts P, Q and R. In which of these does fertilisation occur? **3**



13. Match the organelles with their functions by filling the table: **3**

Organelle	Function
Lysosome	(a) _____
Vacuole	(b) _____
Plastids	(c) _____

14. Explain how chlorofluorocarbons (CFCs) deplete the ozone layer. State two harmful effects of increased ultraviolet radiation on living organisms. **3**

15. A botanist found three plants in a forest. Plant X has no vascular tissue and grows in a damp, shady place. Plant Y has vascular tissue but reproduces by spores and does not produce seeds. Plant Z produces naked seeds in cones and has needle-like leaves.

(a) To which group does plant X belong? 1

(b) To which group does plant Y belong? 1

(c) Name the group of plant Z and give one example. State one more feature of the group. 2

OR

(c) Why are bryophytes called the 'amphibians of the plant kingdom'?

16. Describe the structure of an animal cell with a neat labelled diagram showing at least five parts. State the function of any two parts. **5**

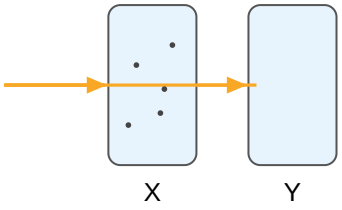
OR

(a) Define diffusion and osmosis. (b) Distinguish between diffusion and osmosis on any three points. (c) Name the process by which a raisin swells in water.

SECTION B — CHEMISTRY — 11 questions · 25 marks

Q.17–19 MCQs; Q.20 Assertion–Reason (options as in Section A).

17. Light is passed through two test tubes X and Y. The path of light is visible in X but not in Y. X and Y could respectively be: **1**



(a) copper sulphate solution and milk (b) milk and salt solution

(c) salt solution and sugar solution (d) sugar solution and salt solution

18. The number of atoms present in one molecule of sulphuric acid (H_2SO_4) is: **1**

(a) 3 (b) 4 (c) 7 (d) 6

19. Which of the following is a pair of isotopes? **1**

(a) ^{14}C and ^{14}N (b) ^{35}Cl and ^{37}Cl (c) ^{40}Ar and ^{40}Ca (d) Na and K

20. **Assertion (A):** Noble gases are chemically inert. **1**
Reason (R): Noble gases are present in the atmosphere.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

21. Two beakers contain (i) chalk powder in water and (ii) salt in water. A torch beam is passed through each. In which beaker is the beam visible? How can chalk be separated from the water? **2**

22. What is meant by formula unit mass? Calculate it for sodium chloride (Na = 23 u, Cl = 35.5 u). **2**

23. Calculate the number of protons and neutrons in $^{37}_{17}Cl$. Why do isotopes have similar chemical properties? **2**

24. 15 g of salt is dissolved in 85 g of water at a temperature where the solubility of salt is 36 g per 100 g of water. (i) Find the % m/m of the solution. (ii) How much more salt can be dissolved in this water to saturate it? **3**

25. Explain the Tyndall effect. Give two daily-life examples. **3**

OR

State three differences between a compound and a mixture.

26. A laboratory technician has to separate the following mixtures.

Mixture	Components
1	Cream and milk
2	Different dyes in an ink

3	Ethanol (b.p. 78 °C) and water (b.p. 100 °C)
4	Common salt and ammonium chloride

(a) Name the technique suitable for mixture 1. 1

(b) Name the technique suitable for mixture 3. 1

(c) Name the technique for mixture 2 and state its principle. 2

OR

(c) Name the technique for mixture 4 and state the principle.

27. Describe an activity to show that a true solution does not scatter light while a colloid does. Write the materials, procedure, observations and conclusion. **5**

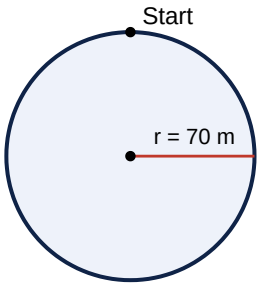
OR

Explain how you would prepare 250 g of a 10 % (m/m) sugar solution. Calculate the masses of sugar and water required and write the steps.

SECTION C — PHYSICS — 12 questions · 23 marks

Q.28–33 MCQs; Q.34 Assertion–Reason (options as in Section A).

28. A cyclist completes one round of a circular track of radius 70 m. Taking π = 22/7, the distance covered and the displacement are respectively: **1**



(a) 440 m, 0 (b) 0, 440 m (c) 220 m, 140 m (d) 440 m, 140 m

29. A body is thrown vertically upwards with a velocity of 20 m s^{-1} (g = 10 m s^{-2}). The maximum height reached is: **1**

(a) 10 m (b) 20 m (c) 40 m (d) 200 m

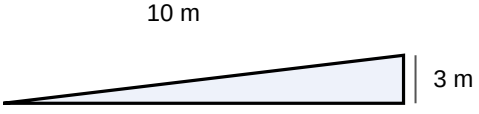
30. A 5 kg block on a smooth surface is acted on by a constant force of 10 N for 3 s, starting from rest. Its velocity after 3 s is: **1**

(a) 2 m s^{-1} (b) 6 m s^{-1} (c) 15 m s^{-1} (d) 30 m s^{-1}

31. A gun of mass 4 kg fires a bullet of mass 50 g with a velocity of 200 m s^{-1}. The recoil velocity of the gun is: **1**

(a) 0.25 m s^{-1} (b) 2.5 m s^{-1} (c) 25 m s^{-1} (d) 800 m s^{-1}

32. The inclined plane shown is 10 m long and 3 m high. Its ideal mechanical advantage is: **1**



(a) 3 (b) 3.33 (c) 0.3 (d) 7

33. The time period of a sound wave of frequency 50 Hz is: **1**

(a) 0.2 s (b) 0.02 s (c) 50 s (d) 0.5 s

34. **Assertion (A):** A body moving with constant velocity has zero acceleration. **1**
Reason (R): Acceleration is the rate of change of speed only, not of direction.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

35. A cricket ball of mass 0.15 kg moving at 30 m s^{-1} is stopped by a fielder in 0.03 s. Calculate (i) the change in momentum and (ii) the average force exerted by the ball on the fielder's hands. **2**

36. The speed of a body is doubled. How does its kinetic energy change? Give a reason. **2**

OR

Define power. Write its SI unit and define it.

37. A train 150 m long crosses a pole in 10 s moving at uniform speed. Find (i) the speed of the train and (ii) the time taken to cross a bridge 300 m long. **3**

38. Priya, a 50 kg student, climbs a flight of stairs 6 m high in 20 s in her school building.

(a) Calculate the gain in her potential energy (g = 10 m s^{-2}). 1

(b) Find the power she develops. 1

(c) Explain why no work is done by gravity on Priya when she walks on a level corridor. 2

OR

(c) State the energy conversion when a ball falls freely from a height.

39. A train starting from rest accelerates at 0.5 m s^{-2} for 1 minute, runs at constant speed for 2 minutes and then decelerates uniformly to rest in 30 s. Find (i) the maximum speed, (ii) the distance in the first minute, (iii) the distance at constant speed, (iv) the retardation and (v) the total distance covered. **5**

OR

Derive the relation $v^2 - u^2 = 2as$ for a body moving with uniform acceleration, using the velocity–time graph.